

46 Animal Reproduction

KEY CONCEPTS

- 46.1** Both asexual and sexual reproduction occur in the animal kingdom *p. 1020*
- 46.2** Fertilization depends on mechanisms that bring together sperm and eggs of the same species *p. 1022*
- 46.3** Reproductive organs produce and transport gametes *p. 1025*
- 46.4** The interplay of tropic and sex hormones regulates reproduction in mammals *p. 1030*
- 46.5** In placental mammals, an embryo develops fully within the mother's uterus *p. 1034*

Study Tip

Make a table: To help you keep track of the roles of the tropic hormones in mammalian reproductive systems, make a table like the one shown here. Complete the last cell for GNRH, and fill in the rows for FSH and LH.

Hormone	Source	In males: Target/effect	In females: Target/effect
GNRH= gonado- tropin- releasing hormone	Hypo- thalamus	Anterior pituitary/ promote FSH and LH release	
FSH=			
LH=			

➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 46
- Animation: Human Spermatogenesis
- Animation: Human Oogenesis

For Instructors to Assign (in Item Library)

- Scientific Skills Exercise: Making Inferences and Designing an Experiment
- Tutorial: Sex Hormones and Mammalian Reproduction



Figure 46.1 This colony of coral polyps is reproducing. Tiny yellow orbs packed with eggs and sperm emerge from the polyps, rise to the sea surface, and burst. There, the eggs and sperm form embryos that become larvae, eventually drifting down and establishing new coral colonies.

In what different ways do animals reproduce?

